

Multimodal LLMs

EE 296 FOUNDATION MODELS AND ADVANCES IN AI

Maximilian Lunt, Nishant Tiwari, Simarpal Singh

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Agenda

Maximilian Lunt

MM-LLMs: Recent Advances in MultiModal Large Language Models

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Multimodal Large Language Models: A Survey

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Break

Simarpal Singh

Perception, Reason, Think, and Plan:
A Survey on Large Multimodal Reasoning Models

Yunxin Li^{*}, Zhenyu Liu^{*}, Zitao Li^{*}, Xuanyu Zhang, Zhenran Xu, Xinyu Chen, Haoyuan Shi, Shenyuan Jiang
Xintong Wang, Jifang Wang, Shouzheng Huang, Xinpeng Zhao, Borui Jiang, Lanqing Hong, Longyue Wang
Zhuotao Tian, Baoxing Huai, Wenhan Luo, Weihua Luo, Zheng Zhang, Baotian Hu¹, Min Zhang
Harbin Institute of Technology, Shenzhen

Project: <https://github.com/HITSz-TMG/Awesome-Large-Multimodal-Reasoning-Models>

Paper 1

Maximilian Lunt

MM-LLMs: Recent Advances in MultiModal Large Language Models

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Presentation Roadmap:

1. Introduction to Multimodal LLMs
2. Evolution of MM-LLMs
3. MM-LLM Architecture
 - a. Multimodal Understanding & Generation
4. Training Pipelines
5. State-of-the-Art Models
6. Benchmarks and Performance
7. Future Directions
8. Conclusion

What are Multimodal LLMs (MM-LLMs)?

Why Multimodal LLMs Emerged

Traditional LLMs were designed mainly for **text processing**

Real-world information exists in many forms

- images
- video
- audio
- text

Earlier multimodal systems required

- massive datasets
- expensive computation

Key Idea Behind MM-LLMs

Powerful pretrained **LLMs** with pretrained **models** for other **modalities**

MM-LLMs can:

- understand multimodal inputs
- perform reasoning across modalities
- generate multimodal outputs
- retain abilities such as:
 - zero-shot learning
 - instruction following
 - language reasoning

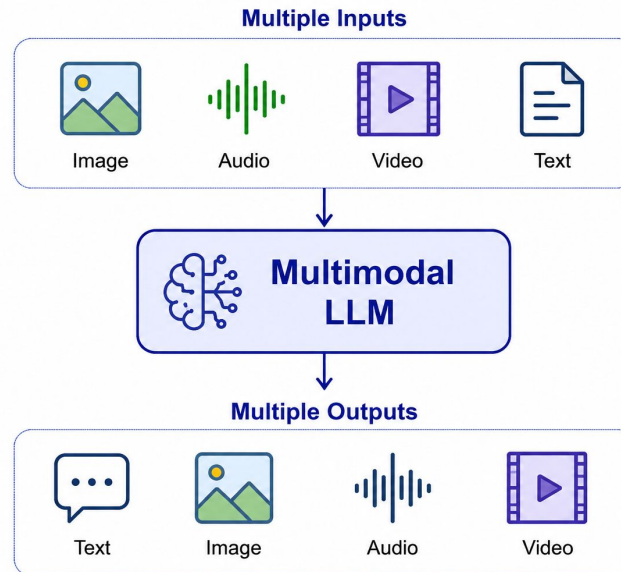


Image generated using ChatGPT

Rapid Expansion of the Field

The field has grown rapidly after systems like:

- GPT-4V
- Gemini
- LLaVA
- MiniGPT-4

Research has expanded from image-text understanding to:

- video understanding
- audio understanding
- multimodal generation
- any-to-any modality systems

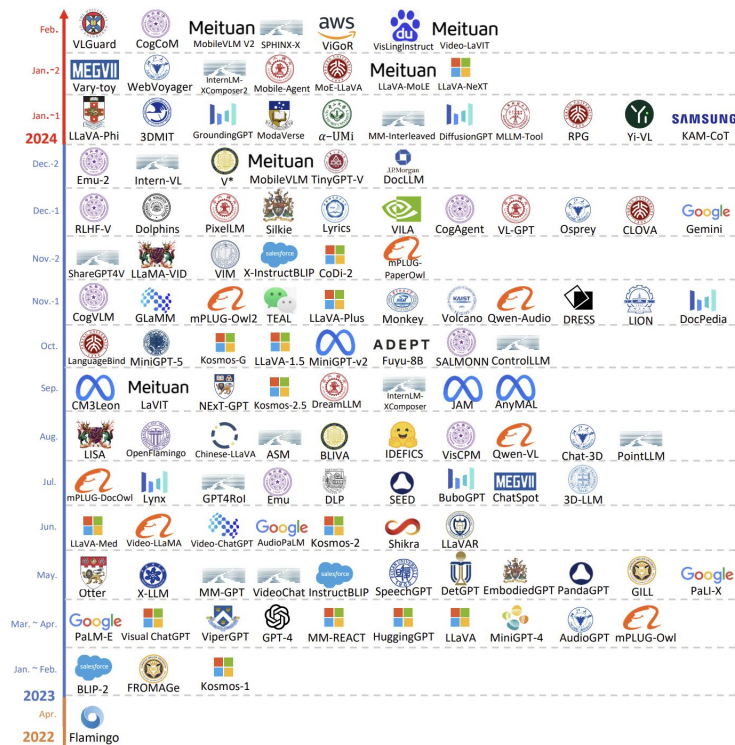


Figure 1: The timeline of MM-LLMs.

General Architecture of MM-LLMs

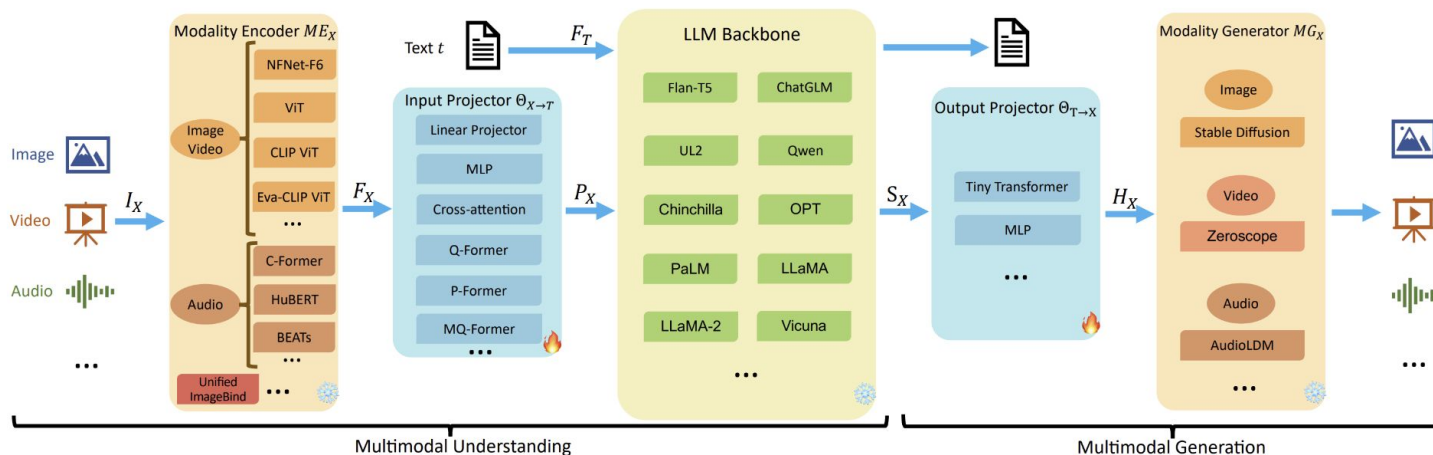


Figure 2: The general model architecture of MM-LLMs and the implementation choices for each component.

MM Understanding

- Images, video, and audio are encoded into feature representations.
- The input projector aligns these features with the LLM text space.

MM Generation

- The LLM reasons over multimodal representations.
- Output projectors and generators produce text, images, audio, or video outputs.

Multimodal Understanding in MM-LLMs

Modality Encoder

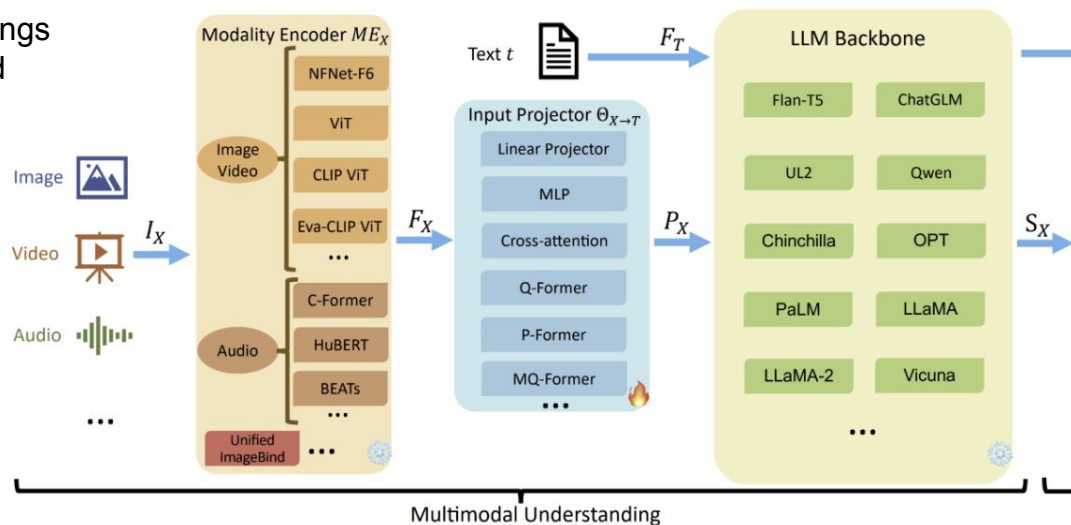
- Images, video, and audio → feature embeddings
- Pretrained encoders: CLIP, ViT, HuBERT, and ImageBind
- Extracts multimodal information

Input Projector

- Aligns features with LLM text space
- Converts encoder outputs for the LLM
- Methods: Linear layers, MLPs, Q-Former

LLM Backbone

- Multimodal reasoning and understanding
- Generates text responses
- Models: LLaMA, Vicuna, Flan-T5, Qwen



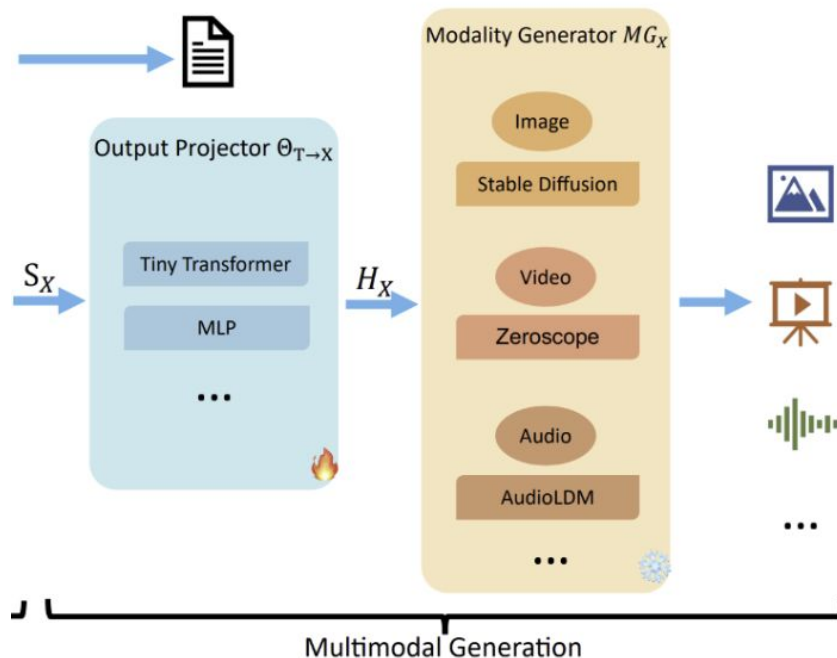
Multimodal Generation in MM-LLMs

Output Projector

- Maps LLM outputs into multimodal features
- Converts representations for generators
- Methods: Tiny Transformers, MLPs

Modality Generator

- Generates images, video, and audio
- Models: Stable Diffusion, Zeroscope, AudioLDM
- Produces multimodal outputs



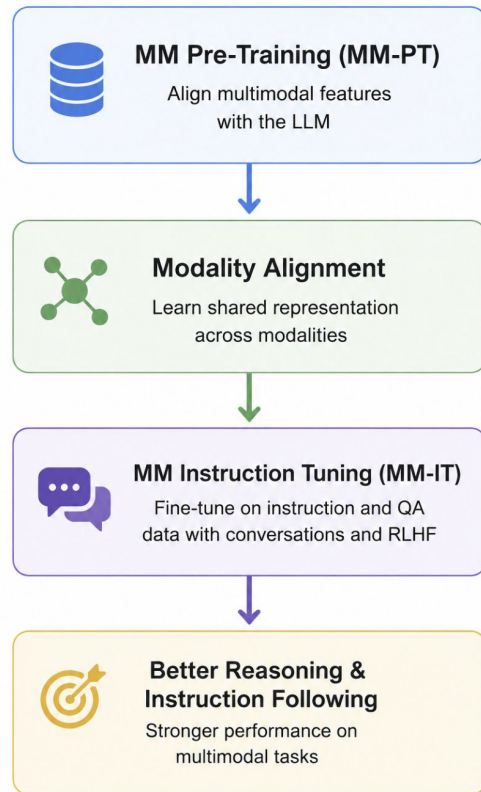
Training Pipeline of MM-LLMs

MM Pre-Training (MM-PT)

- Uses image-text, video-text, and audio-text datasets
- Aligns multimodal features with the LLM
- Trains mainly projector layers
- Enables multimodal understanding

MM Instruction Tuning (MM-IT)

- Fine-tunes pretrained MM-LLMs
- Uses instruction and QA datasets
- Includes conversations and RLHF
- Improves reasoning and instruction following





State-of-the-Art MM-LLMs

Multimodal Understanding

- Understand multimodal inputs
- Generate text outputs
- Tasks:
 - Image captioning
 - VQA
 - Video/audio understanding
- Examples:
 - BLIP-2
 - LLaVA
 - MiniGPT-4

Multimodal Generation

- Generate images, audio, and video
- Uses output projectors + generators
- Supports multimodal content creation
- Examples:
 - NExT-GPT
 - AudioGPT
 - Stable Diffusion systems

Any-to-Any MM-LLMs

- Flexible multimodal inputs and outputs
- Unified multimodal reasoning
- Major future direction of MM-LLMs
- Examples
 - GPT-4V
 - Gemini

Benchmarks and Performance

Model	LLM Backbone	OKVQA	IconVQA	VQA ^{v2}	GQA	VizWiz	SQA ^I	VQA ^T	POPE	MME ^P	MME ^C	MMB	MMB ^{CN}	SEED ^I	LLaVA ^W	MM-Vet	QBench	HM	VSR
Flamingo	Chinchilla-7B	44.7	—	—	—	28.8	—	—	—	—	—	—	—	—	—	—	—	57.0	31.8
BLIP-2	Flan-T5 _{XXL} (13B)	45.9	40.6	65.0	44.7	19.6	61.0	42.5	85.3	1293.8	290.0	—	—	46.4	38.1	22.4	—	53.7	50.9
LLaVA	Vicuna-13B	54.4	43.0	—	41.3	—	—	38.9	—	—	—	—	—	—	—	—	—	—	51.2
MiniGPT-4	Vicuna-13B	37.5	37.6	—	30.8	—	—	19.4	—	—	—	—	—	—	—	—	—	—	41.6
InstructBLIP	Vicuna-7B	—	—	—	49.2	34.5	60.5	50.1	—	—	—	36.0	23.7	53.4	60.9	26.2	56.7	—	—
InstructBLIP	Vicuna-13B	—	44.8	—	49.5	33.4	63.1	50.7	78.9	1212.8	291.8	—	—	—	58.2	25.6	—	57.5	52.1
Shikra	Vicuna-13B	47.2	—	77.4*	—	—	—	—	—	—	—	58.8	—	—	—	—	54.7	—	—
IDEFICS-9B	LLaMA-7B	—	—	50.9	38.4	35.5	—	25.9	—	—	—	48.2	25.2	—	—	—	—	—	—
IDEFICS-80B	LLaMA-65B	—	—	60.0	45.2	36.0	—	30.9	—	—	—	54.5	38.1	—	—	—	—	—	—
Qwen-VL	Qwen-7B	—	—	78.8*	59.3*	35.2	67.1	63.8	—	—	—	38.2	7.4	56.3	—	—	59.4	—	—
Qwen-VL-Chat	Qwen-7B	—	—	78.2*	57.5*	38.9	68.2	61.5	—	1487.5	360.7	60.6	56.7	58.2	—	—	—	—	—
LLaVA-1.5	Vicuna-1.5-7B	—	—	78.5*	62.0*	50.0	66.8	58.2	85.9	1510.7	316.1 [‡]	64.3	58.3	58.6	63.4	30.5	58.7	—	—
+ShareGPT4V	Vicuna-1.5-7B	—	—	80.6	—	57.2	68.4	—	—	1567.4	376.4	68.8	62.2	69.7	72.6	37.6	63.4	—	—
LLaVA-1.5	Vicuna-1.5-13B	—	—	80.0*	63.3*	53.6	71.6	61.3	85.9	1531.3	295.4 [‡]	67.7	63.6	61.6	70.7	35.4	62.1	—	—
MiniGPT-v2	LLaMA-2-Chat-7B	56.9	47.7	—	60.3	30.3	—	51.9	—	—	—	—	—	—	—	—	—	58.2	60.6
MiniGPT-v2-Chat	LLaMA-2-Chat-7B	55.9	49.4	—	58.8	42.4	—	52.3	—	—	—	—	—	—	—	—	—	59.5	63.3
VILA-7B	LLaMA-2-7B	—	—	79.9*	62.3*	57.8	68.2	64.4	85.5	1533.0	—	68.9	61.7	61.1	69.7	34.9	—	—	—
VILA-13B	LLaMA-2-13B	—	—	80.8*	63.3*	60.6	73.7	66.6	84.2	1570.1	—	70.3	64.3	62.8	73.0	38.8	—	—	—
+ShareGPT4V	LLaMA-2-13B	—	—	80.6*	63.2*	62.4	73.1	65.3	84.8	1556.5	—	70.8	65.4	61.4	78.4	45.7	—	—	—

Table 2: Comparison of mainstream MM-LLMs on 18 VL benchmarks. The **red** denotes the highest result, and the **blue** denotes the second highest result. [‡] indicates ShareGPT4V’s (Chen et al., 2023f) re-implemented test results, which are missed in benchmarks or origin papers. * indicates that training images are observed during training.



Future Directions of MM-LLMs

More General Multimodal Systems

- Support additional modalities
- Unified multimodal reasoning and generation
- Move toward fully general AI systems

Better Benchmarks

- Current benchmarks becoming saturated
- Need harder reasoning and real-world tasks
- Improve evaluation for multimodal understanding

Lightweight & Efficient Models

- Reduce computational cost
- Enable deployment on edge and mobile devices
- Improve training efficiency

Embodied & Interactive AI

- Integration with robotics and agents
- Real-world interaction and decision making
- Multimodal perception + action

Reducing Hallucinations & Bias

- Improve reliability of generated outputs
- Reduce incorrect multimodal reasoning
- Address fairness, bias, and privacy concerns



Conclusion

Key Takeaways

- MM-LLMs extend traditional LLMs beyond text into:
 - images
 - video
 - audio
 - multimodal reasoning

Most MM-LLMs use

- pretrained encoders
- lightweight projector layers
- powerful LLM backbones

Training pipelines combine

- multimodal pre-training
- instruction tuning
- RLHF

Major Trends

- Rapid progress after GPT-4V and Gemini
- Shift toward unified any-to-any systems
- Improved multimodal reasoning and generation

Final Insight

- MM-LLMs are moving toward more general, efficient, and human-like AI systems capable of understanding and generating across multiple modalities



References

1. D. Zhang, Y. Yu, J. Dong, C. Li, D. Su, C. Chu, and D. Yu, “MM-LLMs: Recent Advances in Multimodal Large Language Models,” arXiv preprint arXiv:2401.13601, 2024.

Questions?

Thank you!

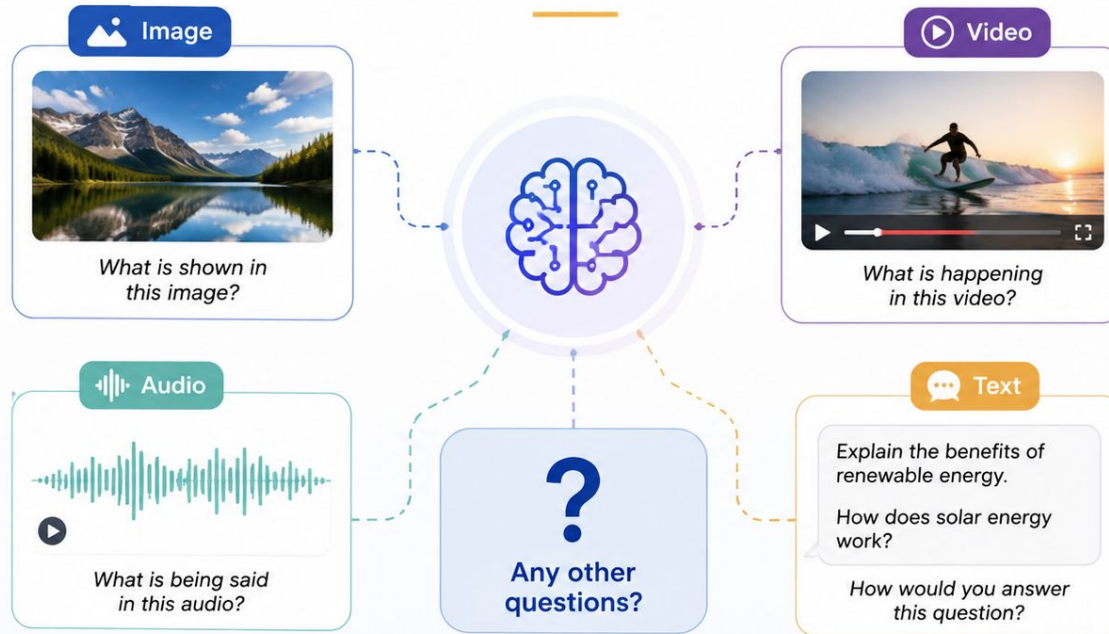


Image generated using ChatGPT



Paper 2

Nishant Tiwari

Multimodal Large Language Models: A Survey

Jiayang Wu¹, Wensheng Gan^{1,2*}, Zefeng Chen¹, Shicheng Wan³, Philip S. Yu⁴

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Presentation Roadmap:

1. **What is multimodal AI?**
2. **How the field evolved**
3. **Technical building blocks**
4. **Important multimodal models**
5. **Applications and datasets**
6. **Challenges and future directions**

Defining Multimodal?

Multimodal learning means using multiple types of data — such as text, images, audio, and video — to understand information more completely.

Homogeneous (Same type of data)

Two cameras capturing the same scene

Heterogeneous (Different types of data)

Images paired with natural language descriptions

Multimodal Data (A combination of data types)

Images, text, audio, video, graphs

Why it matters:

- Humans understand the world using multiple senses
- Multimodal AI can handle richer real-world tasks than text-only models
- Useful in NLP, sentiment analysis, biomedical research, image search, robotics, and human-computer interaction

Historical Timeline of Multimodal Research

1980 - 2000

Single Modality (one data type at a time)

Speech recognition using Hidden Markov Models

Face recognition using PCA

Speech-to-text systems from Dragon Systems

2000 - 2010

Modality Conversion (processing human communication)

AMI: analyzed meeting audio, video, and text

CALO: early intelligent assistant, before Siri

SSP: studied facial expressions, gestures, and voice tone

2010 - 2020

Modality Fusion (deep learning + multimodal data)

Deep Boltzmann Machines: learned cross-modal relationships

Image captioning with attention: describes images using text

2020 - Present

Large-Scale Models (language + perception)

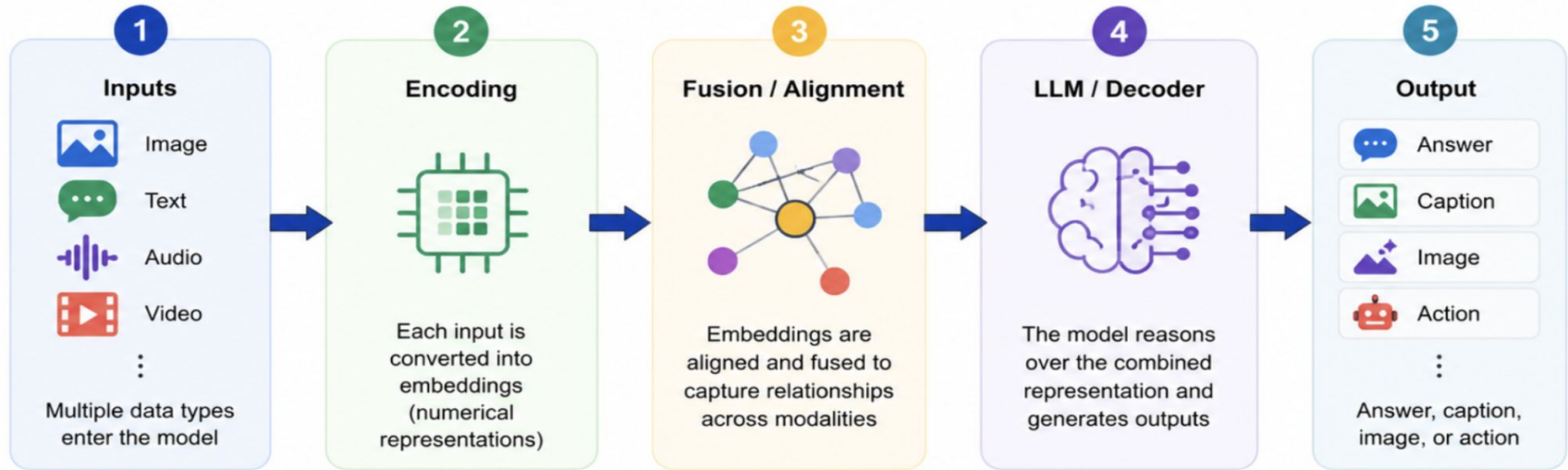
CLIP: image-text alignment

DALL-E: text-to-image generation

PaLM-E: language + vision + robotics

How a Multimodal Model Works:

Multimodal models combine information from different data types to understand and generate outputs



Key Idea: By combining multiple modalities, the model can understand context more completely and perform real-world tasks.

Technical Pillars — Part 1

How Models are Built:

The model must represent different data types, learn useful connections, and choose the right structure.

1. **Knowledge Representation:** Convert text and images into tokens and embeddings
 - a. Text → words/subwords
 - b. Images → regions, grids, or patches
 - c. Goal: make different data types usable by the model
2. **Learning Objectives:** Teach the model to connect modalities
 - a. ITC: match image-text pairs
 - b. MLM/MVM: predict masked text or image parts
 - c. ITM: decide if image and text match
3. **Model Construction**
 - a. Encoder-only: good for retrieval
 - b. Encoder-decoder: good for generation
 - c. Decoder-only: good for autoregressive output

*ITM = Image-Text Matching, MLM = Masked Language Modeling, MVM = Masked Visual Modeling

Technical Pillars – Part 2

4. Information Fusion: How modalities interact

- Fusion encoder: deeper interaction between image/text
- Dual encoder: faster retrieval using separate encoders
- Example:** connects visual objects with words, like matching “dog” in text to the dog in an image

5. Prompt Usage: Guide the model without fully retraining it

- Prompts help adapt models to tasks
- Useful for zero-shot or few-shot tasks
- Example: asking CLIP to classify an image using text prompts

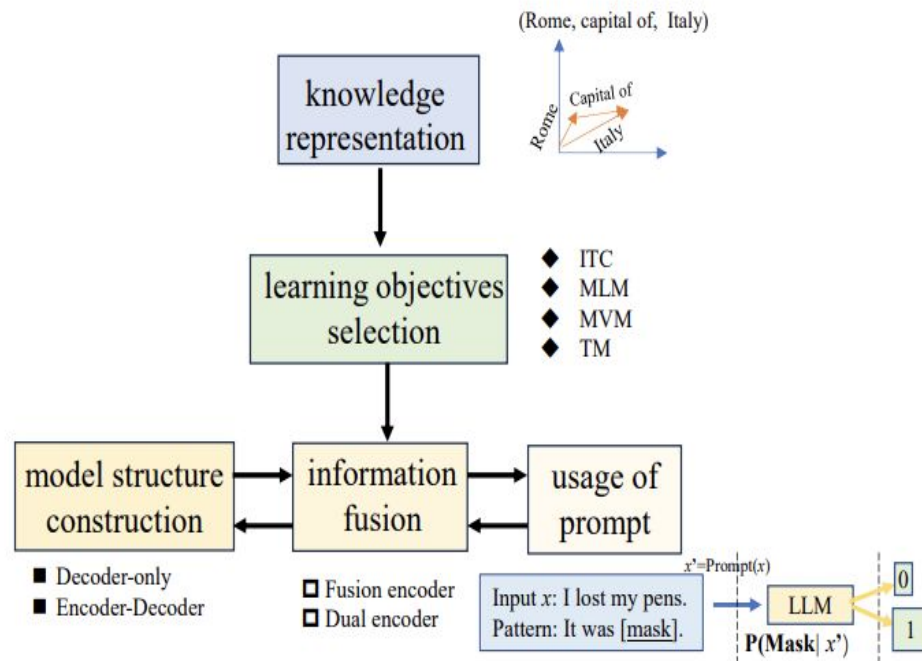


Fig. 3: The technical points of multimodal models.



Foundational Models Behind Multimodal AI

1. **Transformer – 2017:** Introduced attention-based learning for language and sequence tasks
 - a. Uses self-attention to focus on important tokens
 - b. Became the backbone of modern LLMs
 - c. Later adapted for multimodal models
2. **Vision Transformer — 2020:** Applies Transformer ideas to images
 - a. Splits images into small patches
 - b. Treats patches like tokens
 - c. Helped bridge computer vision and NLP
3. **BEiT — 2021:** Applies BERT-style masked learning to images
 - a. Masks image patches during training
 - b. Learns visual representations without heavy labels
 - c. Supports large-scale vision pretraining

Large Scale Multimodal Models (2021-2023)

These models connect language models with visual understanding.

Model	Year	Key Innovation	Why It Matters
Frozen	2021	Frozen LLM + trainable visual encoder	Shows LLMs can support multimodal few-shot learning
Visual ChatGPT	2023	Connects ChatGPT with visual foundation models	Allows image-based conversations and instructions
BLIP-2	2023	Bridges frozen vision encoder and frozen LLM	Efficient image-to-text generation
MiniGPT-4	2023	Uses BLIP-2 + Vicuna with light tuning	Shows GPT-4-like visual-language abilities
LLaVA	2023	Uses instruction tuning for vision-language tasks	Strong multimodal chat and reasoning

Visual Question Answering (VQA) Models

Main idea: Convert visual information into a language-friendly format.

All three models address the same challenge: how can a language model answer questions about images without expensive end-to-end retraining? Each model finds a different way to convert visual information into something an LLM can use.

PICa (2022)

- Uses GPT-3 for VQA
- Converts image → caption → prompt
- Limitation: loses visual details

PNP-VQA (2022)

- Selects image regions related to the question
- Generates captions for relevant patches
- Uses a QA model for zero-shot answering

Img2LLM (2022)

- Converts visual information into question-answer pairs
- Uses those pairs as in-context examples
- No end-to-end training required

Image → visual info → caption / Q&A pairs → LLM answer



Applications of Multimodal Models

1. **Image Captioning:** Image $\rightarrow$ Text
 - a. Generates descriptions for images
 - b. Helps visually impaired users understand images
 - c. Supports image search and tagging
2. **Text-to-Image:** Text $\rightarrow$ Image
 - a. Generates images from text prompts
 - b. Used in digital art and design
 - c. Example: DALL-E 2, Imagen
3. **Sign Language Recognition:** Video/Gesture $\rightarrow$ Text
 - a. Converts gestures into language
 - b. Aligns gestures/video with language or audio
 - c. Useful for accessibility



More Applications of Multimodal Models

4. Emotion Recognition: Audio + text + video + sensors

- a. Detects emotions from multiple signals
- b. Example: music emotion recognition
- c. Uses datasets like DEAM

5. Video Processing: Video + audio + text

- a. Video captioning and video Q&A
- b. Audio-visual speech recognition
- c. Text-to-music generation, like MusicLM

6. Digital Humans (AIGC): Text + vision + speech + animation

- a. 3D avatars and virtual assistants
- b. Lip sync and facial animation
- c. More natural human-computer interaction



Examples of Multimodal Datasets

Multimodal learning depends on datasets that connect multiple data types together.

Dataset	Year	Scale	Modalities
COCO	2014	567K	Image + Text
Visual Genome	2017	5.4M	Image + Text
YouCook2	2018	2.2K	Video + Text
Common Voice	2019	9.2K	Audio + Text
WebVid2M	2021	2.5M	Video + Text
M5 Product	2021	6M	Image + Text + Video + Audio



Key Challenges in Multimodal AI

Challenge	Simple Meaning	Example
Modality Expansion	Models need to handle more than image + text	CT, MRI, PET, sensor data
Training Efficiency	Large models are expensive to train	Distributed clusters, fault tolerance
Continual Learning	Models should keep learning over time	Remember old tasks while learning new ones
Catastrophic Forgetting	New training can damage old abilities	A model may lose prior language/vision skills



My Evaluation of the Paper

Strengths:

1. Gives a broad overview of multimodal LLMs
2. Explains history, technical points, models, datasets, and applications
3. Helpful for beginners entering the field

Limitations:

1. Some model explanations are high-level
2. Survey covers many models but does not deeply compare performance
3. Field changes quickly, so newer models may not be included

Overall: The paper is useful as a survey and introduction, but less useful for detailed model performance comparison.



Key Takeaways and Future Directions

1. Multimodal models address the limitations of text-only LLMs. They combine text, images, audio, video, and other data types for richer understanding.
2. The field has evolved in clear stages: From single-modality systems → modality conversion → modality fusion → large-scale multimodal models.
3. The main open challenges remain:
 - a. Modality expansion
 - b. Training cost and efficiency
 - c. Continual learning
 - d. Catastrophic forgetting

Multimodal AI is a major step toward more general AI, but scaling it reliably remains an open research problem.



Reference:

1. J. Wu, W. Gan, Z. Chen, S. Wan, and P. S. Yu, “Multimodal Large Language Models: A Survey,” *IEEE International Conference on Big Data*, pp. 1–10, 2023.



Questions?

Thank you!





Paper 3

Simarpal Singh

Perception, Reason, Think, and Plan: A Survey on Large Multimodal Reasoning Models

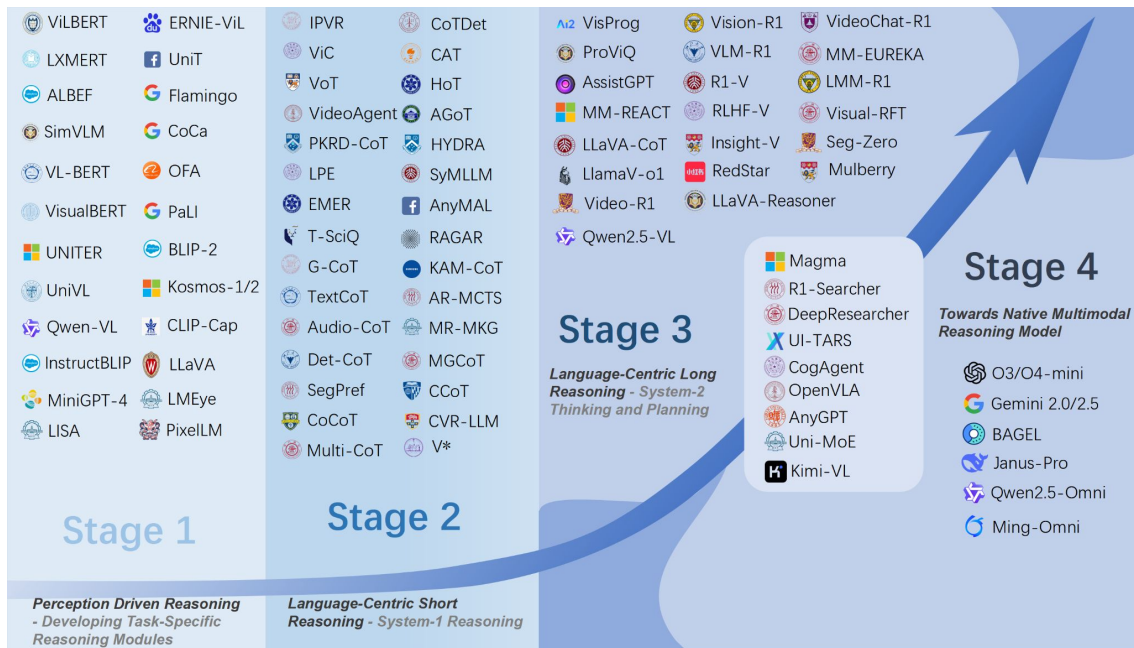
Yunxin Li*, Zhenyu Liu*, Zitao Li*, Xuanyu Zhang, Zhenran Xu, Xinyu Chen, Haoyuan Shi, Shenyuan Jiang
Xintong Wang, Jifang Wang, Shouzheng Huang, Xinpeng Zhao, Borui Jiang, Lanqing Hong, Longyue Wang
Zhuotao Tian, Baoxing Huai, Wenhan Luo, Weihua Luo, Zheng Zhang, Baotian Hu[†], Min Zhang

Harbin Institute of Technology, Shenzhen

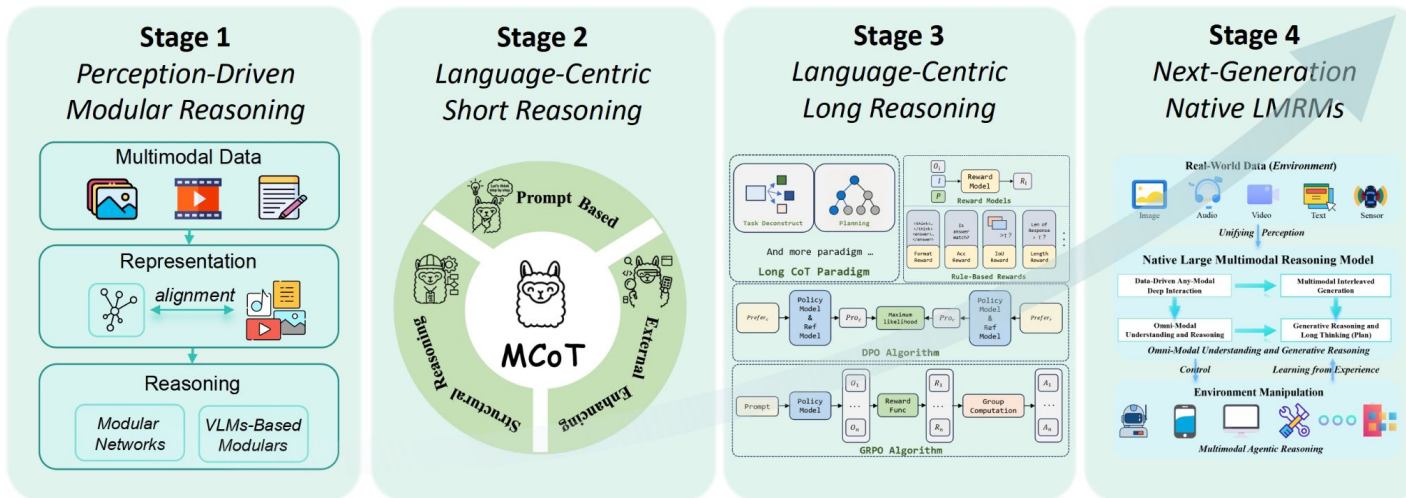
Project: <https://github.com/HITsz-TMG/Awesome-Large-Multimodal-Reasoning-Models>

The Roadmap of LMRNs

Evolving from Modular Pipelines to Native Intelligence

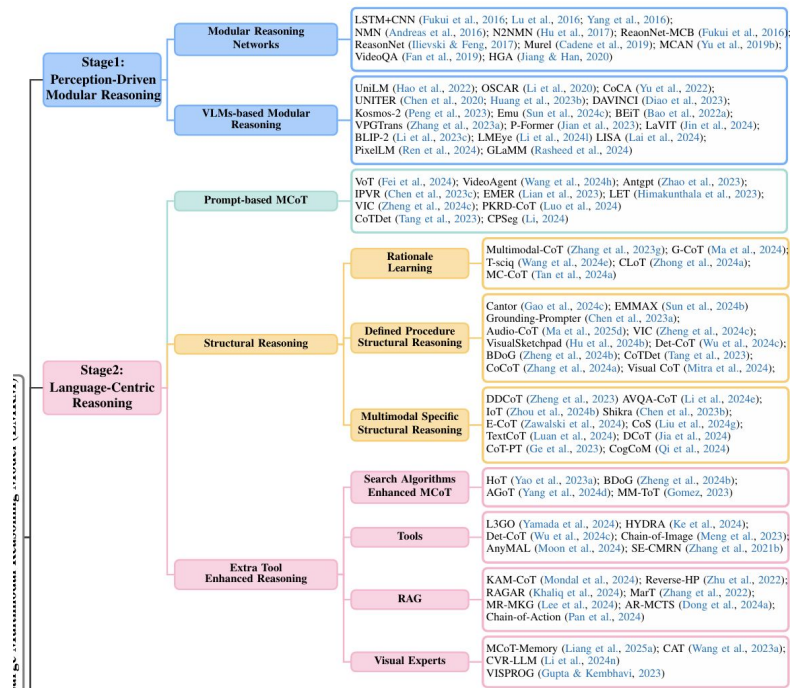


Four Developmental Stages



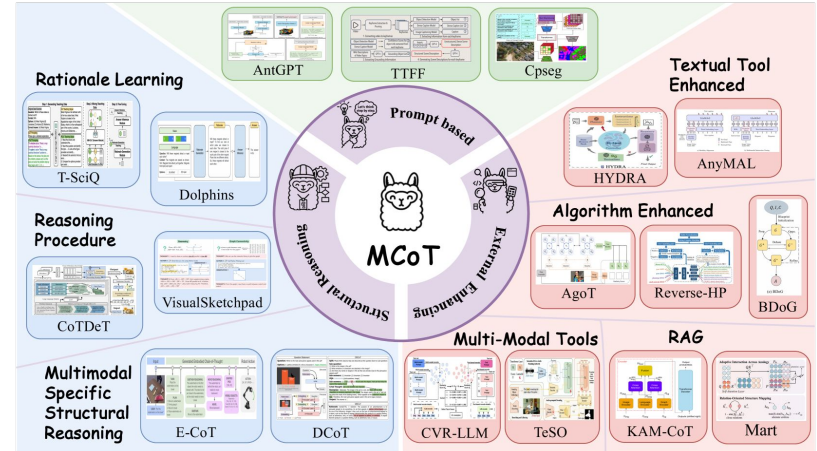
Stage 1: Perception-Driven Modular Reasoning

- **Stitched Multi-Encoder Stack:** Separate visual and text encoders mapping independent modalities into a frozen vector alignment space (e.g., CLIP)
- **Implicit Latent Fusion:** Cross-modal interaction happens entirely inside black-box neural layer computations; no human-readable reasoning trace is output.
- **Downstream Bottleneck:** Requires task-specific head fine-tuning for retrieval or classification, preventing zero-shot generalization.



Stage 2: Language-Centric Short Reasoning - System 1 Reasoning

- **Frozen LLM Core (Late Fusion):** Visual embeddings are projected directly into a frozen text model's token space using linear layers or cross-attention matrices (e.g., LLaVA, BLIP-2).
- **System-1 Autoregressive Tokens:** Reasoning is reactive and continuous. The model generates its rationale tokens on the fly without a computation pause block.
- **External Expert Orchestration:** Uses prompt engineering to route hard problems to external vector databases (RAG), search APIs, or localized visual tools.



Stage 3: Language-Centric Long Reasoning - System 2 Reasoning

- **Test-Time Compute Scaling:** Shifts the computational burden from training-time pre-training to inference-time execution, allowing models to think for long durations before responding.
- **Fine-Grained Verification Rewards:** Optimizes policy models via GRPO or DPO algorithms using programmatic rule-based rewards (Format matching, Accuracy validation, IoU bounding box scoring).
- **Self-Correction Rollouts:** Generates multi-path rollout thinking trees (`<think>...</think>`), allowing the model to detect internal hallucinations and correct its mathematical or spatial orientation vectors mid-stream.

Reasoning Performance on Benchmarks

OmniMMI (Streaming)

Gemini-1.5 & GPT-4o accuracy
< **20%** in complex contexts.

WorldSense (AVQA)

Claude 3.5 achieves only **35%**
accuracy on collaborative
audio-video.

BrowseComp (Agentic)

GPT-4o achieves only **1.9%**
accuracy with browsing tools.

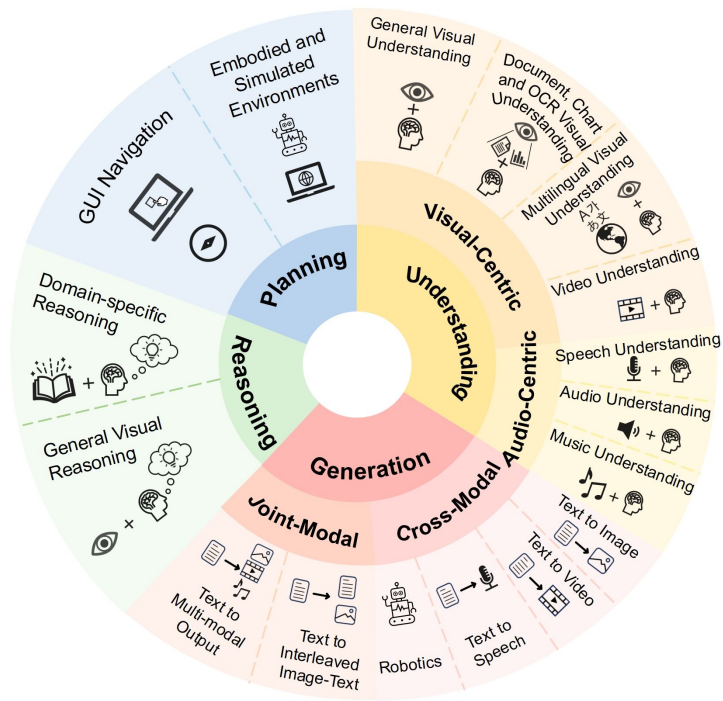
Why they fail: Stitching discrete text backbones to vision encoders creates a fragile translation layer.

The Planning Collapse: Current System-1 and System-2 models lack a grounded, native understanding of environment physics, causing multi-step agent actions to fail catastrophically (1.9% success rate on browsing tasks).

Stage 4: Towards Naive LMRMs

Unified Omni-Modality

- **Early Omni-Tokenization:** Abandons downstream projection matrices. Text tokens, visual pixel coordinates, and audio waveform vectors are projected into a singular, shared embedding space from layer zero.
- **End-to-End Native Fusion:** Cross-modal reasoning is baked directly into every intermediate multi-head attention block of the network.
- **Interleaved Multi-Modal Outputs:** The architecture natively routes generative thinking directly back into action coordinates, synthetic images, or continuous speech streams without switching heads.





Multimodal Agentic Reasoning

Dynamic Adaptation

Adjusting strategies in real-time
based on environmental
feedback (e.g. UI manipulation).

Embodied Learning

Learning through simulated or
physical interactions (e.g.,
OpenVLA, Magma).

Multimodal Agentic Reasoning

Ability	Task	Benchmark	Dataset
	General Visual Reasoning	NaturalBench (Li et al., 2024b), VCR (Zellers et al., 2019) PhysBench (Chow et al., 2025b), MMBench (Liu et al., 2024) MMMU (Yue et al., 2024), AGIEval (Zhong et al., 2024b) MMStar (Chen et al., 2024), InfographicVQA (Mathew et al., 2022) VCRBench (Qi et al., 2025), VisualPuzzles (Song et al., 2025) IV-Bench (Ma et al., 2025a), VisualLogic (Xu et al., 2025) FG-BM3V (Yu et al., 2025a), Video-MMLU (Song et al., 2025a) DVBench (Zeng et al., 2025), GeoSense (Xia et al., 2025c) FLIP Plesner et al. (2025), VILBench (Tu et al., 2025b) HAVEN (Gao et al., 2025a), MAGIC-VQA (Yang et al., 2025d) PM4Bench (Gao et al., 2025b), FAVOR-Bench (Tu et al., 2025) VLEMBenc (Ruan et al., 2025), UrbanVideo-Bench (Zhao et al., 2025a) FortisAVQA (Ma et al., 2025b), VideoVista-CulturalLingo (Chen et al., 2025b) CoMT (Cheng et al., 2025b), CCLiell (Zhong et al., 2025b) Wethink-dataset (Yang et al., 2025b), MMMG (Yao et al., 2025c) Do You See Me (Kanade & Ganu, 2025), SpaCE-10 (Gong et al., 2025) GaslightingBench-R (Zhu et al., 2025), MultiNet (Guruprasad et al., 2025) MANBench (Zhou et al., 2025a), VCR-SFT (Wang et al., 2025b) TempVS benchmark (Song et al., 2025d), SFE benchmark (Zhou et al., 2025d) HSSBench (Kang et al., 2025), Argus Inspection (Yao et al., 2025d) ChenTable (Zhou et al., 2025c), KoshuMD-10 (Lin et al., 2025b)	VCR (Zellers et al., 2019), TDLUC (Kaffe & Kanan, 2017) MMPR (Wang et al., 2024g), ChartQA (Masry et al., 2022) SWAG (Zellers et al., 2018), LLaVA-CoT (Xu et al., 2024b) CLEVR (Johnson et al., 2016), Mulberry-260K (Yao et al., 2024a) ShareGPT4oReasoning (Zhang et al., 2024c), R1-Onevision (Yang et al., 2025f) Video-R1-data (Feng et al., 2025b), Visual-CoT (Shao et al., 2024)
		MathVista (Lu et al., 2024), MATH-Vision (Wang et al., 2024c) VLM-Bench (Zheng et al., 2022), GemBench (Garcia et al., 2024) GeoQA (Chen et al., 2022a), VIMA-Bench (Jiang et al., 2022) WorldSimBench (Qin et al., 2024), WorldModelBench (Li et al., 2025c) ScienceQA (Lu et al., 2022), ChartQA (Masry et al., 2022) MathQA (Amini et al., 2019), Habitat (Savva et al., 2019) A12-THOR (Kolve et al., 2017), Gibson (Xia et al., 2018) iGibson (Li et al., 2021a), Isaac Lab (Mittal et al., 2023) VCBench (Wang et al., 2025w), VCT (Götting et al., 2025) 3MDBench (Sviridov et al., 2025), PuzzleBench (Zhang et al., 2025g) ColorBench (Liang et al., 2025b), VisualPuzzles (Song et al., 2025f) ProcXMLE (Cui et al., 2025), NoTeS-Bank (Pal et al., 2025) EIBench (Lin et al., 2025b), XLR5-Bench (Wang et al., 2025c) STI-Bench (Li et al., 2025b), EgoToM (Li et al., 2025p) DomainQA (Zhong et al., 2025b), MM-HQ (Cui et al., 2025) MMCR-Bench (Yan et al., 2025a), Misleading ChartQA (Chen et al., 2025m) FlowVerse (Chen et al., 2025a), VisNumBench (Weng et al., 2025) MicroVQA (Burgess et al., 2025), MPBench (Xu et al., 2025b) Open3DVQA (Zhan et al., 2025), ProKBench (Yang et al., 2025) Chart-HQA (Chen et al., 2025b), MMSciBench (Ye et al., 2025b) CharXiv (Wang et al., 2024e), SPAR-Bench (Zhang et al., 2025f) YSI-Bench (Yang et al., 2025b), STI-Bench (Li et al., 2025f) VRBench (Yu et al., 2025b), MATP-BENCH (He et al., 2025) MMRefine (Paik et al., 2025), MMR-V (Zhu et al., 2025b) WikiMixQA (Perevoutin et al., 2025), SciNet (Wang et al., 2025c) MultiFiBlen (Feng et al., 2025), RealFiBench (Wu et al., 2025d) VideoMathQA (Rasheed et al., 2025), GDI-Bench (Li et al., 2025g) PhysicsArena (Dui et al., 2025b), Spoken-MQA (Wei et al., 2025a) ReasonSeg-Diff (Wang et al., 2025g), VF-Eval (Song et al., 2025c) ChartMind (Wei et al., 2025c)	Habitat (Savva et al., 2019), A12-THOR (Kolve et al., 2017) Gibson (Xia et al., 2018), GeoQA (Chen et al., 2022a) Isaac Lab (Mittal et al., 2023), ProcTHOR (Deitke et al., 2022) CALVIN (Mess et al., 2022), SRMA-SRMEval (Miao et al., 2025) SpaceR-151k (Ouyang et al., 2025)
		MathVista (Lu et al., 2024), MATH-Vision (Wang et al., 2024c) VLM-Bench (Zheng et al., 2022), GemBench (Garcia et al., 2024) GeoQA (Chen et al., 2022a), VIMA-Bench (Jiang et al., 2022) WorldSimBench (Qin et al., 2024), WorldModelBench (Li et al., 2025c) ScienceQA (Lu et al., 2022), ChartQA (Masry et al., 2022) MathQA (Amini et al., 2019), Habitat (Savva et al., 2019) A12-THOR (Kolve et al., 2017), Gibson (Xia et al., 2018) iGibson (Li et al., 2021a), Isaac Lab (Mittal et al., 2023) VCBench (Wang et al., 2025w), VCT (Götting et al., 2025) 3MDBench (Sviridov et al., 2025), PuzzleBench (Zhang et al., 2025g) ColorBench (Liang et al., 2025b), VisualPuzzles (Song et al., 2025f) ProcXMLE (Cui et al., 2025), NoTeS-Bank (Pal et al., 2025) EIBench (Lin et al., 2025b), XLR5-Bench (Wang et al., 2025c) STI-Bench (Li et al., 2025b), EgoToM (Li et al., 2025p) DomainQA (Zhong et al., 2025b), MM-HQ (Cui et al., 2025) MMCR-Bench (Yan et al., 2025a), Misleading ChartQA (Chen et al., 2025m) FlowVerse (Chen et al., 2025a), VisNumBench (Weng et al., 2025) MicroVQA (Burgess et al., 2025), MPBench (Xu et al., 2025b) Open3DVQA (Zhan et al., 2025), ProKBench (Yang et al., 2025) Chart-HQA (Chen et al., 2025b), MMSciBench (Ye et al., 2025b) CharXiv (Wang et al., 2024e), SPAR-Bench (Zhang et al., 2025f) YSI-Bench (Yang et al., 2025b), STI-Bench (Li et al., 2025f) VRBench (Yu et al., 2025b), MATP-BENCH (He et al., 2025) MMRefine (Paik et al., 2025), MMR-V (Zhu et al., 2025b) WikiMixQA (Perevoutin et al., 2025), SciNet (Wang et al., 2025c) MultiFiBlen (Feng et al., 2025), RealFiBench (Wu et al., 2025d) VideoMathQA (Rasheed et al., 2025), GDI-Bench (Li et al., 2025g) PhysicsArena (Dui et al., 2025b), Spoken-MQA (Wei et al., 2025a) ReasonSeg-Diff (Wang et al., 2025g), VF-Eval (Song et al., 2025c) ChartMind (Wei et al., 2025c)	Habitat (Savva et al., 2019), A12-THOR (Kolve et al., 2017) Gibson (Xia et al., 2018), GeoQA (Chen et al., 2022a) Isaac Lab (Mittal et al., 2023), ProcTHOR (Deitke et al., 2022) CALVIN (Mess et al., 2022), SRMA-SRMEval (Miao et al., 2025) SpaceR-151k (Ouyang et al., 2025)
Multimodal Planning	GUI Navigation	WebArena (Zhou et al., 2024c), Mind2Web (Deng et al., 2023) VisualWebBench (Liu et al., 2024b), OSWorld (Xie et al., 2024a) OmniACT (Kanyov et al., 2024), VisualAgentBench (Li et al., 2024d) LlamaTouch (Zhang et al., 2024d), Windows Agent Arena (Bonatti et al., 2024) Feret-UI (You et al., 2024), WebShop (Yao et al., 2022) SWE-bench M (Yang et al., 2024c), MineDojo (Fan et al., 2022) TeamCraft (Long et al., 2024), VMAGE (Zhong et al., 2025d) TongUI (Zhang et al., 2025a), BEARCLUBS (Song et al., 2025e) MCA-Bench (Wu et al., 2025b), ThinkGeo (Shabbir et al., 2025)	AMEX (Chai et al., 2024), RiCo (Deka et al., 2017) WebSRC (Chen et al., 2021), E-ANT (Wang et al., 2024d) Androsifw (Toyama et al., 2021), GUI-World (Chen et al., 2024b) MBE-ARI (Noronha et al., 2025)
		MineDojo (Fan et al., 2022), MineDojo (Fan et al., 2022) GVCCL (Kim et al., 2023), BEHAVIOR-1K (Li et al., 2024c) Habitat 3.0 (Puig et al., 2024), SAPIEN (Xiang et al., 2020) HomeRobot (Yenamandra et al., 2023), HoloAssist (Wang et al., 2023b) DrivingDojo (Rietsch et al., 2022), WolfBench (Qiao et al., 2024) MBE-ARI (Noronha et al., 2025), Vi4scape (Lim et al., 2025) AttachSecure-Bench (Wang et al., 2025a)	MineDojo (Fan et al., 2022), Habitat 3.0 (Puig et al., 2024) SAPIEN (Xiang et al., 2020), HomeRobot (Yenamandra et al., 2023) HoloAssist (Wang et al., 2023b), DrivingDojo (Rietsch et al., 2022) OmniHD-Scenes (Zheng et al., 2025c)



Technical Prospects of N-LMRMs

- **Unified Cross-Modal MoE:** Implementing Mixture-of-Experts to route multi-modal data traffic to dedicated experts, preventing visual token swelling from degrading linguistic performance.
- **Natively Interleaved Long CoT:** Scaling test-time compute by allowing the model to natively execute visual crop, zoom, or sensor filtering commands *mid-thought*.
- **Automated Cross-Modal Data Loops:** Engineering synthesis pipelines that algorithmically align text, video, and localized sensor maps to generate pristine pre-training data.



My Insights

Severe Data Bottleneck

Training Efficiency

Reality of System-2
Reasoning



Thank You

Questions?